



Humanism, posthumanism, and compassionate love

Stephen G. Post*

Center for Medical Humanities, Compassionate Care, and Bioethics, HSC Level 3, Rm. 080, Stony Brook University, Stony Brook, New York 11794-8036, USA

ABSTRACT

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This article considers a special area where pure scientism, separated from humanistic perspectives, has come to define a perspective broadly termed posthumanism, where the modification of human nature is too easily embraced as progress. The humanistic response to this scientism is equally central in this reflection. I offer a perspective on the futility of pure scientism with regard to human progress worthy of our dignity as human creatures. I conclude with an endorsement of the natural human capacity for compassionate love, for it is still the discovery of what already lies within us that dignifies what lies before us.

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1. Introduction

In 1959, C.P. Snow described “a gulf of mutual incomprehension” between scientists on the one hand and literary intellectuals or humanists on the other. This gulf remains wide today, making deeper progress in our understanding of many questions about nature, human nature, and the cosmos more difficult. Yet when progress on such questions occurs, it is usually synergistic rather than monistic. Robert Frost wrote, “Something there is that doesn’t love a wall” [1]. Tremendous intellectual generativity exists in those areas where deep integration has occurred, such as in the study of empathy and altruism, mind and brain, neuroscience and ethics, science and religion, positive psychology, and bioethics. Still, there is a countervailing tendency, especially among academicians, toward the safe silos of sharp disciplinary boundaries. Frost also wrote, “Good fences make good neighbors” [1]. Let us hope that these are only waist-high picket fences that allow for good rapport in the neighborhood. Scientists and humanists in a synergistic *rapprochement*, with exemplary integrative thinkers at the table, can do what neither can achieve alone.

The gulf described by Snow is evident around questions of the malleability of human nature. In such contexts humanists can help to deepen more technocratic and “too-easy” scientific visions of so-called posthumanists, while scientists can bring an enthusiasm for innovation that certainly should be appreciated even when it appears shallow. Ideally, with the right forums, support, and intellectual leadership available to allow a genuine face-to-face rapport, in-group/out-group tribalism can give way to synergy and wisdom. Wisdom requires a rapport and *rapprochement* that allows for deeper probing of complex issues.

The posthumanist is the self-proclaimed scientific visionary who views human dignity in large part as a matter of seizing the opportunity to modify and enhance the species in fundamental ways including the deceleration or arrest of aging, genetic engineering, the man/machine symbiosis of nanotechnology and cybernetics, robotics, reproductive cloning, and even the downloading of mind into immortalizing computers. Biotechnology and informational technology converge in an exhilaration that is contagious. Enter the anti-posthumanist, who is also the pro-humanist, asserting that our dignity lies chiefly in accepting the existing contours of human nature as the fine-tuned gift of evolutionary processes, and that biotechnological

* Tel.: +1 631 444 9797; fax: +1 631 444 9744.

E-mail address: stephen.post@stonybrook.edu

efforts beyond the therapeutic intended to recreate that nature are arrogant and short-sighted impositions on future generations. The critics of posthumanism add, with their own enthusiasm, that no technological modifications can provide genuine purpose, happiness, wisdom, compassionate love, and similar virtues that make for a good life. Nor can such posthumanist dreams address the needs of the neediest, where global justice comes into play.

In this article I consider a special area where pure scientism, separate from humanistic perspectives, has come to define a perspective broadly termed posthumanism, where the modification of human nature is too easily embraced as progress [2, 3]. The humanistic response to this scientism is equally central in this reflection [4]. I will offer a perspective on the futility of a pure scientism with regard to human progress worthy of our dignity as human creatures. As a rather conserving humanist [5], I conclude with an endorsement of the natural human capacity for compassionate love, for it is the discovery of what already lies within us that dignifies what lies before us.

2. Two perspectives on the future of human nature

The posthumanist, inculcated in a culture of scientism, is a self-proclaimed manipulator of nature and of human nature. I had a colleague in bioengineering who gave the most exhilarating and glossy Powerpoint lectures to his impressionable undergraduate students, leaving them assured that there is no holding up “progress” toward the new human. A wonderfully creative mind and a terrific scientist, my colleague did not have much forbearance when a student sheepishly asked, “But maybe we should let well enough alone?” There was a boldness in my colleague’s delivery, a kind of in-your-face dismissal of doubters for being backward. He was highly entrepreneurial and ambitious institutionally. He was pure scientism on wheels, defiantly ready to politicize critics as “right wing conservatives.” I admired my colleague’s wit and dramatic flare. But there was nothing probing or self-critical in his words. He was ideological in that any facts or opinions that challenged posthumanism were ignored. Some posthumanists, of course, are more deeply reflective than others.

On the other side of the spectrum, away from the culture of scientism and in the domain of thoughtful humanism, is the always impressive Professor Leon R. Kass of the University of Chicago. I had the opportunity as a graduate student to respond to a presentation on the pitfalls of posthumanism and “enhancement” medicine by Professor Kass at Chicago, where I was serving in 1983 as a teaching assistant in a course entitled “Social Issues in Medicine.” At the time, Professor Kass was writing his classic work, *Toward a More Natural Science* [6], still among the most influential and important books in the field of ethics and medicine. A man knowledgeable in the sciences and medicine, his intellectual roots lie equally in the great philosophical classics, especially of antiquity. His lectures were for me always a delight because he could draw from the vignette of almost any Greek writer and show how helpful it could be in gaining a critical perspective on the hubris of posthumanism. In comparison with my bioengineering colleague, Professor Kass was surely the deeper thinker, resisting the posthumanist exhilaration over a brave new world. But Kass was not so much an anti-posthumanist as he was a pro-humanist, asserting that our dignity lies chiefly in accepting the existing contours of human nature as the fine-tuned gift of evolutionary processes. Biotechnological efforts beyond the therapeutic, intended to recreate that nature, are arrogant and short-sighted impositions on future generations that fail to appreciate the gift is what we are. It was regrettably not surprising to me that so many professional bioethicists—a term of opprobrium to some—reacted so strongly to Professor Kass’s interest in such themes as human dignity and excellence while he was chairing the Bioethics Commission during the years of George W. Bush’s presidency [7]. In contrast, I agree with Professor Kass that no technological modifications can provide us with genuine purpose, happiness, wisdom, compassionate love, and similar virtues that make for a good life. Nor can such posthuman dreams address the needs of the neediest around the globe.

Let me proceed, then, with further elaboration of the posthumanism versus humanism discussion, which is essentially to contrast scientism with a deeper tradition of reflection.

3. The culture of posthumanism

Posthumanists tell us that boundaries have already been crossed and that there is no road back. Their sense of the inevitability of the new human is hardly convincing, but this is part of a wonderful enthusiasm in their passion for innovation that seems to be such a core aspect of human creativity. The vast changes predicted elicit media swirl and best-selling books with terrifically colorful titles. Two of the more distinguished titles are Gregory Stock’s *Metaman: The Merging of Human and Machines into a Global Superorganism* [8], and Lee Silver’s *Remaking Eden: Cloning and Beyond in a Brave New World* [9]. Stock is a Hopkins-trained biophysicist with a Harvard MBA; Silver is professor of molecular biology at Princeton. These biologists are matched by visionaries of computer science and engineering, such as Ray Kurzweil, author of *The Age of Spiritual Machines: When Computers Exceed Human Intelligence* [10].

However provocative and difficult to envision, these books present plausible visions that place human beings as we know them on the endangered species list. These ideas of freedom from biological constraint and of taking on our natural role as the new co-creators of a new nature have a certain attraction. The 1974 Nobel Laureate Christian de Duve [11] thoughtfully urges us to pursue the goal of a “superorganism” as we “reshape life,” as he raises the question: “After us, what?” Posthumanists do not believe that biology should in any sense be destiny, as they seek to engineer a new sort of entity for whom human nature has been more or less overcome [12]. We are to take human nature into our own re-creative hands as the next great step in evolution, achieving our full creative maturity. And to do so, argued the theologian Karl Rahner, is to understand what it means to be made in the image of God. After all, humans have been reinventing themselves anyway through applied

technologies for millennia, for we are a technological species. As the Princeton University physicist Freeman Dyson writes, “The artificial improvement of human beings will come, one way or another, whether we like it or not,” as scientific understanding increases, for such improvement has always been viewed as a “liberation from past constraints” [13].

Was not the very idea of human beings trying to fly deemed heretical hubris in the light of eternity – *sub specie aeternitatis*. Let us set aside trepidation and with confidence rethink ourselves in the light of human creativity. The postmodernists have paved the way by purportedly demonstrating that there is no essential aspect to human nature, so *vive le difference*. Long live deconstructionism! Stock’s second book, entitled *Redesigning Humans: Our Inevitable Genetics Future*, introduces the idea of “superbiology” as we take full control of our own biology in turning toward perfection [14]. Posthumans are the advanced models that we humans today will design for tomorrow, and as such, they are no longer humans as we know the term.

It is noteworthy that the posthumanist emphasis is on the biotechnological enhancement of discrete human units, the single one. This is in tension with the humanistic perfectibilism we associate not with biotechnology, but rather with Dewey, Rousseau, Comte, Kropotkin, Sorokin, the Dalai Lama, and those who have stressed human enhancement through the strengthening of prosocial bonds, virtues, and the capacity for sympathy or compassion. Contemporary evolutionary biology and group selection theory make this social view of perfection more valid than any atomistic one.

From whence cometh this atomistic rage for techno-enhancement? David F. Noble [15] has argued with some plausibility that the roots of this posthumanist project lie in Western religion, and especially in the ninth century, when the “useful arts” came to be associated with the concept of human redemption. As a result, we have a “religion of technology” that gives rise to an uncritical and irrational affirmation of unregulated technological advance, which is too easily deemed good. Noble hopes that we can free ourselves from the religion of technology through learning to think and act rationally toward humane goals.

Others have noted the religious element. As Gerald Gruman pointed out in his definitive entry in the *Dictionary of the History of Ideas* [16], the modern concern with enhancing longevity “stems from the decline since the Renaissance of faith in supernatural salvation from death; concern with the worth of individual identity and experience shifted from an other-worldly realm to the ‘here and now,’ with intensification of earthly expectations” (16, p. 88). Thus did George Bernard Shaw, in his remarkable *Back to Methuselah* (1921) take up the variations in life spans between species such as parrots and dogs, or turtles and wasps, and asked that science rise to the theologically redemptive occasion by vastly increasing human longevity [17]. We now have the serious search for an anti-aging pill [18], and a host of researchers working on caloric restriction, anti-oxidation, and the basic science of aging with the goal of slowing the aging process [19,20,21].

Indeed, the Renaissance Christian humanist Francis Bacon, a founder of the scientific method, in his millennialist and utopian work, *The New Atlantis* (1627), set in motion a biological mandate for boldness that included both the making of new species or “chimeras,” organ replacement, and the “water of Paradise” that would allow the possibility to “indeed live very long” [22]. This tradition of biological ambition has been well traced by S.J. Olshansky [23,24].

We stand at the threshold and must ask ourselves, how brave a new world is still a good world? And here arise the humanists who are anti-posthumanist, which is to say they repudiate the pure scientism of those who would too easily remake humanity. The theme of human hubris and technology is of course perennial [25], but now more relevant than ever [26].

4. The culture of humanism

For every utopian posthumanist there is a dystopian humanist. In 1727, a century after Bacon’s *The New Atlantis*, Jonathan Swift portrayed in *Gulliver’s Travels* a group of immortals overwhelmed by severe dementia [27,28], raising the specter of a scientific victory over finitude that would result in a sad protraction of morbidity. Let us glance at these perennial debates over anti-aging as an example.

Should the future individual, viewing his or her own prospects for deceleration of aging, pursue such anti-aging treatments when and if they actually become available? Perhaps yes, if this assures one that diseases for which old age is the overwhelmingly significant risk factor can be avoided. But there is an important school of thought that cautions against the development of treatments to slow aging. One hears, in the posthumanist anti-aging mantras, the sounds of financial ambition, a touch of hubris, and a conviction that all of this change is inexorable. Certainly, as a first point of critique, start-up companies abound. There is a strong entrepreneurial edge to posthumanism, and it is an investor’s dream. This fact should make us nervous [29].

Then there is that nagging matter of injustice. The voices of the great masses of humanity, in the meanwhile, cry out for food and water, for sanitation and housing. In the light of worldwide injustice and poverty, posthumanist thinking is viewed by some as nearly obscene. Posthumanism is the plaything of those who can afford it. On the other hand, this has been the case with so much human progress. Technologies are first available at high cost to the wealthy, but eventually become reasonably available to all.

What of human flourishing? The individual, when confronted with the availability of decelerated aging, ought to reflect carefully about the choice at hand, raising every question of relevance to self and to humanity. One of the wiser minds of the last century, Hans Jonas (d. 1993), an intellectual inspiration for today’s anti-posthumanists, articulated these questions quite thoroughly. He wrote in 1985 that “a practical hope is held out by certain advances in cell biology to prolong, perhaps indefinitely extend, the span of life by counteracting biochemical processes of aging” [30, p. 18]. How desirable would this power to slow or arrest aging be for the individual and for the species? Do we want to tamper with the delicate biological “balance of death and procreation” [30, p. 18], and preempt the place of youth? Would the species gain or lose? Jonas, by

merely raising these questions, meant to cast significant doubt on the anti-aging enterprise. “Perhaps,” he wrote, “a nonnegotiable limit to our expected time is necessary for each of us as the incentive to number our days and make them count” [30, p. 18]. Jonas’s later essays raising many of these same questions were published posthumously [31].

These questions are echoed in the writings of Kass, who accepts biotechnological progress within a therapeutic mode; rather, his issue is chiefly with efforts to reshape human nature. In a creative essay, “L’Chaim and Its Limits: Why Not Immortality?” [32], Kass provides arguments against prolonging life. He asserts, for example, that the gradual descent into aged frailty weans us from attachment to life and renders death more acceptable; that our numbered days encourage a creative depth in our humanity, a depth that escaped so many of the immortal Greek gods and goddesses whose often debauched and purposeless behavior made Plato wish to ban from the ideal *Republic*; that a preoccupation with the continuance of our lives is a distraction from that which is best for our souls; that in a world transformed by anti-aging research, youth will be displaced rather than elevated, and the parental investment in the young will give way to “my” perpetuation; and that in such a new world we will grow bored and tired of life, having “been there” and “done that.” There is wisdom in simply accepting the fact that we evolved for reproductive success rather than for long-lived lives. Without such wisdom will we lose sight of our deepest creative motives?

And what about the politics of the new and old model humans? The technological dystopians, such as Aldous Huxley, come to mind here. So do the writings of C.S. Lewis. An early anti-posthumanist, Lewis wrote *The Abolition of Man* in 1944 [33] in part as a response to all that unfolded in German science in the 1930s. Lewis defended a natural-law tradition: what is good, and we should live within certain limits. He cautioned against a world in which one class of enhanced human beings would dominate and oppress the other. We might ask, then, if those freed from the decline of aging would become the superior and elite humans, while those who age are deemed inferior.

A leading anti-posthumanist, Francis Fukuyama, is the author of the widely reviewed book entitled *Our Posthuman Future: Consequences of the Biotechnology Revolution* [34]. The ambitions of posthumanists to create a new posthuman who is no longer human are, he asserts, lacking in fundamental appreciation for natural human dignity. He argues that the anti-aging technologies of the future will disrupt all the delicate demographic balances between the young and the old, and exacerbate the gap between the haves and the have-nots.

The anti-posthumanists will often make appeal to nature and character as morally valuable categories. They understand the proper human attitude toward our evolved nature as one of humility, awe, and appreciation. They exhort us to work with our human nature to get the best out of it, rather than to seek cavalier dominion in an effort to recreate what is already good. And what is best in human nature? The most universal claim, readily articulated in all worthy spiritual and moral traditions over the centuries, is compassionate love. So let us turn to this theme in conclusion.

5. Compassionate love and the human good

We are bombarded by offers to enhance us as human beings. Botox, anabolic steroids, human growth hormone to make our children a little taller, and the dubious promises of a fountain of youth—all are for sale but none add to dignity or a deeper happiness. Let us instead focus our attention not on the external vessel of our bodies but on the capacity for a generous love that already lies within us waiting only to be more fully unveiled and engaged. Without this love, we are poor even if we do not know it. With this love we grow in dignity and happiness [35,36]. *In the giving of self lies the discovery of a deeper and happier self.* Here lies the perennial truth that unites all the great religions. And here lies the inner core of human enhancement and dignity, which has nothing to do with biotechnological tinkering.

There is, I believe, a more important way of thinking about the kind of human creatures we are than the posthumanists have allowed. Our dignity rests not in gadgets and biotechnological wizardry. Rather, our dignity as human beings is already ours to claim when we treat another person with love and justice, and when we manifest virtues as varied as perseverance, faith, courage, forgiveness, gratitude, commitment, humility, integrity, and hope. The expressions of the ways of love, from celebration and attentive listening, to creativity and helping, to loyalty and respect, are all at the very heart of human dignity. Loyal friendship, a concern for the needy, and the worry we have over a wayward child who is not necessarily our own are what make us creatures of dignity and grace.

Granted, we are less acculturated to speaking of dignity than we are of rights, autonomy, and equality, but we need such an idea. Nothing else will do. Etymologically, the English word derives from the Latin *dignitas*, meaning honor, elevation, and worthiness. This dignity does not flow from “future shock” opportunities to modify and enhance human nature in fundamental ways. Dignity might best be realized by expanding the modern idea of love from the narrow domain of the nearest and dearest to all humanity, and thereby discovering a politics of love in the mode of Gandhi or King or Farmer. Rather than free ourselves of biological constraints in a misplaced effort to transcend humanness by technology, let us re-emphasize a love that does justice. But let us use every technique of science to better understand how to raise caring children, how to love wisely and effectively, how to harvest the neurological connection between giving and delight.

As for the changes posthumanists seek, new evils may be more perilous than old ones. To use an analogy, we are like the sailor who climbs as high on the mast as one can in order to rise high above the waters of nature, but only to see the boat capsize under our weight tipping the mast into the waves below [25]. How can we presume that the brave new world will be a better world? It will only be better if it is world of love and justice, and these are things that lie within our hearts and minds as currently constituted.

As for the two cultures of science and humanities, humanism must guide science, and restrain it, while maintaining our focus on the ultimate source of human dignity and enhancement: the love that lies within us and that perhaps even shapes the universe [26].

In the art of compassion, much of the tribalism that widens the gulf between Snow's "two cultures" can be ameliorated. Human beings are psychologically oriented to group identity, and easily fall into attitudes of "them" versus "us." This element of evolutionary psychology can be overcome, I believe, when "we" unite in a common effort to solve the most significant risks to the human future. One risk is surely the unwise use of science to modify the contours of nature and of human nature. In confronting such perils, we can perhaps grow in a greater appreciation and love for those who approach these perils from a perspective different than our own.

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References

- [1] Frost R. Mending Wall. In: Untermeyer L, editor. Modern American poetry. [1919]. New York: Bartleby.com. Available from: <<http://www.bartleby.com/104/64.html>>; 1999.
- [2] Kurzweil R. The age of spiritual machines: when computers exceed human intelligence. New York: Viking; 1999.
- [3] Hayles NK. How we become posthuman: virtual bodies in cybernetics, literature and informatics. Chicago: University of Chicago Press; 1999.
- [4] Habermas J. The future of human nature. Cambridge: Polity Press; 2003.
- [5] Kristol W, Cohen E, editors. The future is now: America confronts the new genetics. New York: Rowman & Littlefield; 2002.
- [6] Kass LR. Toward a more natural science: biology and human affairs. New York: Free Press; 1985.
- [7] Kass LR. Available from: <<http://www.bioethics.gov/material/kasspaper.html>>; 2003 [website no longer available].
- [8] Stock G. Metaman: the merging of human and machines into a global superorganism. New York: Simon & Schuster; 1993.
- [9] Silver L. Remaking Eden: cloning and beyond in a brave new world. New York: Avon; 1997.
- [10] Kurzweil R. The age of spiritual machines: when computers exceed human intelligence. New York: Penguin; 1999.
- [11] de Duve C. Life evolving: molecules, mind, and meaning. Oxford: Oxford University Press; 2002.
- [12] Hook CC. Transhumanism and posthumanism. In: Post SG, editor. The encyclopedia of bioethics. 3rd ed., 5 vols.. New York: Macmillan Reference; 2004. p. 2517–20.
- [13] Dyson FJ. Imagined worlds. Cambridge, MA: Harvard University Press; 1997.
- [14] Stock G. Redesigning humans: our inevitable genetic future. Boston: Houghton Mifflin; 2002.
- [15] Noble DF. The religion of technology: the divinity of man and the spirit of invention. New York: Penguin; 1997.
- [16] Gruman GL. Longevity. In: Weiner PP, editor. Dictionary of the history of ideas. New York: Charles Scribners; 1973. p. 88–93.
- [17] Benecke M. In: Rubenstein R, editor. The dream of eternal life: biomedicine, aging, and immortality. New York: Columbia University Press; 2002.
- [18] Lane MA, Ingram DK, Roth GS. The serious search for an anti-aging pill. Scientific American 2002;Aug:36–41.
- [19] Masoro EJ. Caloric restriction's effects on aging: opportunities for research on human implications. Journal of Gerontology 2001;Series A(Special Issue 1):56A.
- [20] Miller RA. Extending life: scientific prospects and political obstacles. Milbank Quarterly 2002;80:155–74.
- [21] National Institute on Aging. Action plan for aging research: strategic plan for fiscal years 2001–2005. Washington, DC: U.S. Department of Health and Human Services; 2001.
- [22] Bacon F. The new Atlantis. In: Vickers B, editor. Francis Bacon: a critical edition of the major works. Oxford: Oxford University Press; 1996. p. 457–89.
- [23] Olshansky SJ, Carnes BA. The quest for immortality: science at the frontiers of aging. New York: W.W. Norton; 2001.
- [24] Olshansky SJ, Hayflick L, Carnes BA. No truth to the fountain of youth. Scientific American 2002;286:92–5.
- [25] Niebuhr R. The nature and destiny of man. New York: Charles Scribners Sons; 1941.
- [26] Post SG, editor. The encyclopedia of bioethics. 3rd ed., 5 Vols. New York: Macmillan; 2004.
- [27] Swift J. Gulliver's travels. Garden City, NY: Doubleday; 1945. 1727 original.
- [28] Post SG. The moral challenge of Alzheimer disease: ethical issues from diagnosis to dying. Baltimore, MD: The Johns Hopkins University Press; 2000.
- [29] Juengst ET, Binstock RH, Mehlman M, Post SG. The social implications of genuine anti-aging interventions: the need for public dialogue. Science 2003; 299:1323.
- [30] Jonas H. The imperative of responsibility: in search of an ethics for the technological age. Chicago: University of Chicago Press; 1985.
- [31] Jonas H. Mortality and morality: a search for the good after Auschwitz. Evanston, IL: Northwestern University Press; 1996.
- [32] Kass LR. L'chaim and its limits: why not immortality? First Things 2001;13:17–24.
- [33] Lewis CS. The abolition of man [1944]. New York: Simon & Schuster; 1996.
- [34] Fukuyama F. Our posthuman future: consequences of the biotechnology revolution. Baltimore, MD: The Johns Hopkins University Press; 2002.
- [35] See <<http://www.unlimitedloveresearch.com>>.
- [36] Post SG. Unlimited love: altruism, compassion, and service. Philadelphia: Templeton Foundation Press; 2003.

Stephen G. Post is Professor of Preventive Medicine and Director of the Center for Medical Humanities, Compassionate Care, and Bioethics at Stony Brook University. He was previously (1988–2008) Professor of Philosophy, Religious Studies, and Family Medicine in the School of Medicine, Case Western Reserve University, and Senior Research Scholar at the Becket Institute of St. Hugh's College, Oxford University. Post is a Senior Fellow in the Center for the Study of Law and Religion at Emory University, and a Distinguished Senior Advisor for the Positive Psychology Center at the University of Pennsylvania. His research has been supported by the National Institutes of Health and the National Endowment for the Humanities. His book, *The Moral Challenge of Alzheimer Disease: Ethical Issues from Diagnosis to Dying* [28] was widely influential. He has contributed numerous studies pertaining to dementia on topics such as genetic testing, anti-aging technologies, and hospice care. In 1994, Post was elected a Hastings Center Fellow, and a Senior Scholar in the Kennedy Institute of Ethics at Georgetown University. Post co-founded the Bioethics Network of Ohio. He served as editor-in-chief of the definitive reference work in the field of bioethics, the third edition of the five-volume *Encyclopedia of Bioethics* [12]. Post is a leader in the study of altruism, compassion, and love in the integrative context of scientific research, health care delivery and outcomes, philosophy, and religious thought.